Here are some helpful YouTube links and online resources to cover the topics in BCA-405 Mathematics – III:

UNIT-I: Complex Variables

1. **YouTube** - *Introduction to Complex Numbers*
   https://www.youtube.com/watch?v=f0qF0AqZ9Uk
2. **YouTube** - *Polar Form of Complex Numbers*
   https://www.youtube.com/watch?v=Tc1-KgGZCkA
3. **YouTube** - *Functions of Complex Variables*
   https://www.youtube.com/watch?v=7Vqu1jfX6Qw
4. **Khan Academy** - *Inverse Trigonometric Functions*
   https://www.khanacademy.org/math/trigonometry/trig-inverse-functions

UNIT-II: Sequence, Series, and Convergence

1. **YouTube** - *Sequences and Series*
   https://www.youtube.com/watch?v=IbZ1G0FLHc0
2. **YouTube** - *Convergence of Sequences*
   https://www.youtube.com/watch?v=kcv27cC8s-E
3. **YouTube** - *Test for Convergence of Series*
   https://www.youtube.com/watch?v=pxN7_-8FymY
4. **Khan Academy** - *Tests for Convergence (Ratio Test, Root Test)*
   https://www.khanacademy.org/math/calculus-1/cs1-series/cs1-convergence-tests/a/convergence-test-review

UNIT-III: Vector Calculus

1. **YouTube** - *Introduction to Vector Calculus*
   https://www.youtube.com/watch?v=ZlLPwD4hbAI
2. **YouTube** - *Gradient, Divergence, and Curl*
   https://www.youtube.com/watch?v=s4fHZ6jmDjc
3. **Khan Academy** - *Vector Fields and the Gradient*
   https://www.khanacademy.org/math/multivariable-calculus/partial-derivatives/multivariable-chain-rule/a/gradient

UNIT-IV: Fourier Series

1. **YouTube** - *Introduction to Fourier Series*
   https://www.youtube.com/watch?v=3Pcy6yOWwJw
2. **YouTube** - *Fourier Series of Even and Odd Functions*
   https://www.youtube.com/watch?v=wQh3l8t-YQ0

3. **YouTube** - *Half Range Fourier Series*
   https://www.youtube.com/watch?v=EmtbxdoF_J8

UNIT-V: Ordinary Differential Equations of First Order

1. **YouTube** - *First Order Differential Equations*
   https://www.youtube.com/watch?v=IbMxb5t0XqQ
2. **Khan Academy** - *Homogeneous Differential Equations*
   https://www.khanacademy.org/math/differential-equations/diffeq-first-order/first-order-homogeneous-differential-equations/a/homogeneous-differential-equations
3. **YouTube** - *Solving Bernoulli's Differential Equation*
   https://www.youtube.com/watch?v=dcXTBzB2RmE

UNIT-VI: Ordinary Differential Equations of Second Order

1. **YouTube** - *Second Order Homogeneous Differential Equations*
   https://www.youtube.com/watch?v=aH0HrHokZxI
2. **YouTube** - *Solving Differential Equations with Constant Coefficients*
   https://www.youtube.com/watch?v=fuSzPxLk9Co
3. **YouTube** - *Cauchy's Differential Equations and Solutions*
   https://www.youtube.com/watch?v=lLOgXQwAcLg
4. **Khan Academy** - *Solving Second-Order Linear Differential Equations*
   https://www.khanacademy.org/math/differential-equations/second-order-differential-equations

These resources should provide both theoretical explanations and practical examples to help you master the topics in **Mathematics – III** for your BCA course.